

Real-Time Retail CPI For Bolivia

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Abstract

Official inflation metrics, while robust, suffer from publication lags due to traditional in-person surveying methods. To address this critical reporting gap, I construct a real-time, synthetic Retail Price Index for Bolivia utilizing daily web-scraped supermarket data from a major national chain. Employing an AI-driven classification pipeline, I map over 28,000 unique products across three major cities to a “core-5” basket aligned with official Instituto Nacional de Estadística (INE) categories. I then use a chained Jevons index to calculate and aggregate daily price relatives. The resulting synthetic index demonstrates a strong correlation with official INE data, achieving coefficients up to 0.99 for index levels and 0.80 for month-over-month inflation. Furthermore, regression analysis reveals that the daily synthetic index serves as a statistically significant leading indicator, with 15-day and 20-day lags successfully predicting overall official inflation. Ultimately, this real-time tracker equips policymakers with an essential, high-frequency tool to monitor price dynamics in real time and preempt future inflationary pressures.

1 Introduction

In the summer of 2025, Bolivia suffered its worst inflation episode in over 40 years. Prices rose by 20% year-over-year in 2025, and real GDP fell by 1.2% on an annual basis [1] [2]. The fall in output was primarily due to large real declines in commerce and mineral extraction [3]. The stagflationary environment contributed to a projected rise in the poverty rate (defined here as the share of the population living below \$4.20 in 2021 PPP terms) from 4.9% in 2024 to 6.4% in 2025 [4]. Bolivians’ suffering translated into a resounding defeat of the incumbent party, Movimiento al Socialismo (MAS), in the 2025 presidential election.

While the government blamed rampant speculation and international inflationary pressures, more plausible explanations include a sharp increase in the money supply resulting from debt monetization, an artificial gas shortage created by subsidies, unanchored inflation expectations, and dollar shortages caused by capital, currency, and export controls. If the newly elected Paz administration wishes to permanently vanquish the specter of a rising price level, it must be equipped with the highest-frequency data possible.

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Although the INE (Instituto Nacional de Estadística) publishes a robust monthly CPI, its methods rely on traditional in-person surveying, creating substantial lags between price observation and publication [1]. Researchers such as Alberto Cavallo and Roberto Rigobon pioneered real-time CPI measurement using web-scraped data from major Latin American supermarkets [5]. In their 2013 paper, “Online and official price indexes: Measuring Argentina’s inflation,” Cavallo and Rigobon laid out a methodology for constructing a real-time CPI. I apply that methodology to a large Bolivian supermarket chain to create a real-time retail CPI for the Bolivian economy.

2 Methodology

2.1 The Supermarket

The supermarket’s prices are collected daily by a scraper coded by Mauricio Foronda and updated live on GitHub [6]. The chain is one of the largest in Bolivia and has more than 30 locations across La Paz, Santa Cruz, and Cochabamba, allowing me to construct city-specific CPIs. The company sells a wide range of products with more than 28,000 unique IDs, and that product universe changes over time as items are added and removed. Product subcategories are summarized in Table 9. Given the retailer’s size and product offerings, the resulting index provides a reasonable estimate of formal supermarket prices.

2.2 Product Categorization

Although the supermarket’s own subcategories provide substantial granularity, I wanted to ensure that each of the more than 28,000 products was mapped into one of the 12 official INE categories using the Classification of Individual Consumption According to Purpose (CCIF). To do so, I used an AI-coded application trained on the official methodology and had it categorize each product individually. The categorizer included additional categories because the CCIF places Food (and Beverages) Consumed Away from Home within Restaurants and Hotels, whereas the INE methodology only includes Food Consumed Away from Home in the CPI. I exclude that category from the final analysis because of its small sample size.

To deal with newly added products, I used an AI-coded program that draws from a persistent historical mapping dictionary (`mappings/Final_Complete_Categories.csv`) to identify known products. When a new product is added, the script recognizes it and queries the Gemini 2.5 Flash model. The model is provided with the official INE 2016 Methodology PDF and the CCIF categorization guidelines as strict context. It is forced to output a JSON payload classifying the new item into one of the designated core categories (or “Unmapped” if inapplicable). The AI’s decisions are appended back to the master CSV, ensuring the API is only queried for novel products.

Table 1: Core-5 Expenditure Categories and Weights

Category	Raw Weight	Normalized Weight (%)
Food and Non-Alcoholic Beverages	27.06	55.08
Clothing and Footwear	7.56	15.39
Miscellaneous Goods and Services	7.55	15.37
Furniture, Household Goods and Services	6.08	12.38
Alcoholic Beverages and Tobacco	0.88	1.79

2.3 The Core Basket

Because the supermarket either had low observation counts in some divisions, such as communications, or clearly did not represent the same basket of goods covered by the official CPI, such as transportation or housing, I defined a “core-5” basket designed to track the INE measure more accurately. This core basket combines high product counts with substantial within-category coverage. For example, although Recreation and Culture contains more than 1,000 unique products, its within-category coverage is still limited because the official INE basket also includes items such as sporting event tickets, movies, and tourist packages.

2.4 Other Filtering

Any product tagged seasonal (temporada) is removed from the sample due to the artificial downward bias it introduces into the index. For example, the Temporada Navideña (Christmas Season) has no listings throughout the year except directly before Christmas, when the prices spike and after which they fall dramatically and are removed. Because my methodology utilizes daily price relatives, when these prices drop and are later removed without a price increase, they bias the index calculation downwards.

2.5 Elementary Jevons Index

The core calculation relies on a chained Jevons index, the standard geometric formula used by Cavallo (2013) for elementary aggregate inflation when product substitution is likely.

Step A: Day-over-Day Relatives

For every active product, the daily relative price change is calculated against its immediately previous observed price:

$$\text{RelativeDOD}_{i,t} = \frac{\text{Price}_{i,t}}{\text{Price}_{i,t-1}}$$

Missing prices are omitted from the daily relative calculation, ensuring that the index only moves

when a price change is actively observed.

Step B: Attrition and Imputation Limit (40-Day Rule)

The pipeline enforces a strict 40-day inclusion limit. If a product is unobserved for more than 40 consecutive days, it is treated as permanently dropped from the matched-model sample. If the exact same product reappears on day 41 or later, its day-of-return relative is nullified. The product then establishes a new base price for the following day, preventing artificial index spikes generated by severely outdated baseline prices while keeping the overall CPI structure intact.

Step C: Elementary Jevons Aggregate

Within each city (c) and category (k), the elementary index for day t is the geometric mean of all constituent product relatives:

$$\text{JevonsMean}_{c,k,t} = \exp \left(\frac{1}{n} \sum_{i=1}^n \ln (\text{RelativeDOD}_{i,t}) \right)$$

Step D: Time-Series Chaining

The daily geometric means are chained sequentially using a cumulative product to construct a continuous time series anchored at 100:

$$\text{ChainedIndex}_{c,k,t} = \text{ChainedIndex}_{c,k,t-1} \times \text{JevonsMean}_{c,k,t}$$

2.6 Aggregation

Once the elementary category indices are calculated, they are aggregated upward to form macroeconomic indicators.

City-Level Aggregation

For a given city, the overall CPI is the weighted arithmetic average of its five core category indices using the normalized weights established earlier in the paper:

$$\text{CityCPI}_{c,t} = \sum_k (\text{ChainedIndex}_{c,k,t} \times \text{NormalizedWeight}_k)$$

National Aggregation

The national CPI is computed as the weighted arithmetic average of the three metropolitan indices using official demographic or economic weights:

- Santa Cruz: 41.47%
- La Paz: 39.10%
- Cochabamba: 19.43%

$$\text{NationalCPI}_t = \sum_c (\text{CityCPI}_{c,t} \times \text{RegionalWeight}_c)$$

Daily Forward-Filling & N-Counts

To keep the tracker continuous, I forward-fill any missing dates with the previous date’s level. The need to employ this method is rare, with only 10 days being forward-filled throughout the entire sample period.

The pipeline also continuously tracks the number of relative price counts by each category to allow users to monitor data degradation and days with low observations. The measure adds price-relative observations across cities, treating each product-city combination as a unique observation, meaning that a single product sold across the three cities will be counted three times, as it provides different pricing information within each city.

Automation Infrastructure

The entire pipeline runs via a GitHub Actions Cron Job at 12:00 UTC (8:00 AM local time) daily. The runner uses a Python 3.10 environment, triggers the AI mapping for novel IDs, runs the calculations, and commits the updated historical .csv and .json artifacts back to the repository.

3 Results

Table 2 reports summary statistics for the full distribution of day-over-day price relatives. Category-level summary statistics are reported in Appendix Table 6.

Table 2: Overall Price Relative Summary Statistics

N	Min	P25	Median	Mean	Geom Mean	P75	Max	SD
21,199,211	0.0101	1	1	1.0007	1.0005	1	108.4722	0.0344

Here we see that the median and mean price relatives remain centered around one, with most observations concentrated at that value. Some products nonetheless display implausibly large one-day jumps. For example, Salsa BBQ Sauce Hot Promos 510 gr (ID: 832253) moves from \$5 at the official exchange rate to \$557 in a single day before later being corrected back to \$5.6. While this event is clearly a reporting glitch, imposing an arbitrary cap or floor on price relatives could introduce bias into the estimates. For instance, if a product priced at \$0.01 were adjusted to \$1

during a high-inflation episode, the implied relative of 100 would likely be removed by any reasonable outlier rule despite being valid.

The geometric average mitigates the influence of these extreme observations by allowing for substitution within categories. Appendix Figure 6 illustrates the geometric mean as a function of a single outlier price relative while all other product prices remain fixed. At a sample size of 6000, roughly the city-average number of food and beverage price relatives, the marginal effect of one outlier is small and declines as the outlier grows. For these reasons, I do not apply a range constraint to “reasonable” price relatives, because doing so could introduce bias.

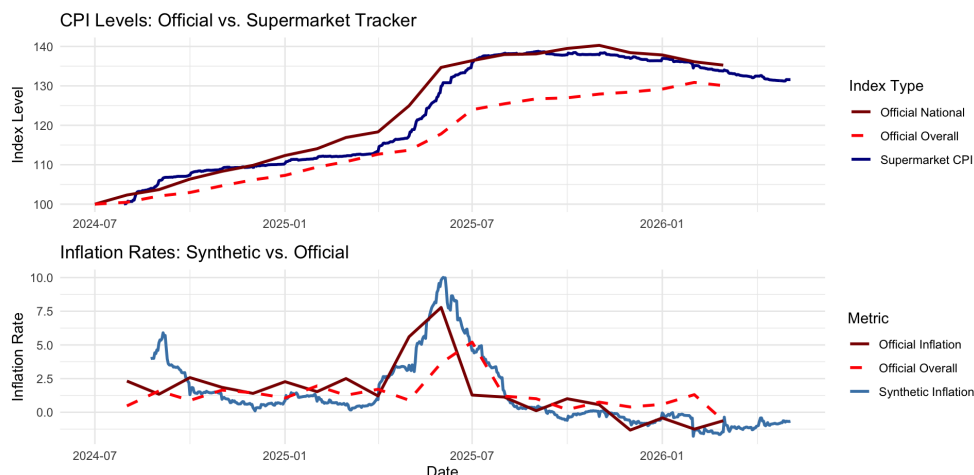


Figure 1: Retail CPI compared with the reweighted official CPI and the overall CPI.

The level and month-over-month inflation estimates for the official data and the synthetic tracker move together with little divergence. Both the “core-5” official CPI and the real-time tracker appear to act as leading indicators for overall CPI. From May to August 2025, the real-time CPI rises before the overall CPI does, implying that information could have been used to inform policy well before overall inflation registered the price acceleration.

CPI levels between the synthetic tracker and official data are highly correlated, with the synthetic national and official national (“core-5”) having a coefficient of 0.99. Official and synthetic Santa Cruz have a coefficient of 0.98, while Cochabamba and La Paz have 0.98 and 0.99, respectively. The synthetic correlation with the official overall CPI (full 12 categories) remains high with a coefficient of 0.97, despite the compositional differences between the two. Inflation rates between the synthetic and official data are also correlated. Synthetic national inflation and official national inflation (“core-5”) have a coefficient of 0.8. For the city-level data, Santa Cruz synthetic inflation and official (“core-5”) inflation have a coefficient of 0.72, while Cochabamba and La Paz have 0.71 and 0.7, respectively. Finally, national synthetic inflation and overall official inflation (full 12 categories) have a coefficient of 0.71.

Overall, the synthetic CPIs tend to stay within 2–3 units of their official counterparts, while synthetic inflation measures remain within 1–2 percentage points of their official equivalents on average. The one exception is that there is a larger average absolute gap (6.04 units) between the national overall (full 12 categories) CPI and the synthetic CPI.

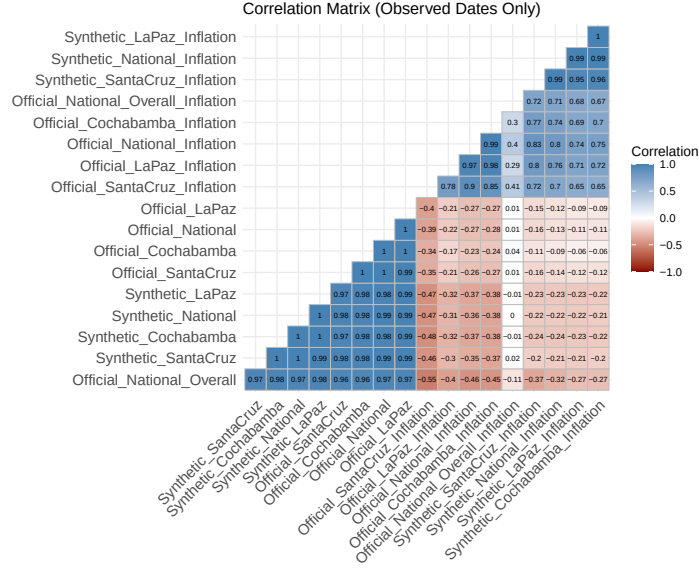


Figure 2: Correlation matrix of national and city-level CPI and inflation for the overall CPI, the equivalent “core-5” CPI, and synthetic inflation estimates

Table 3: Mean Absolute Difference (MAD) between Synthetic and Official CPI

Region	Metric	Mean Absolute Difference
Cochabamba	Index	2.302
La Paz	Index	2.793
National (“core-5”)	Index	2.146
National Overall	Index	6.039
Santa Cruz	Index	2.250
Cochabamba	Inflation	1.470
La Paz	Inflation	1.380
National (“core-5”)	Inflation	1.257
National Overall	Inflation	1.374
Santa Cruz	Inflation	1.534

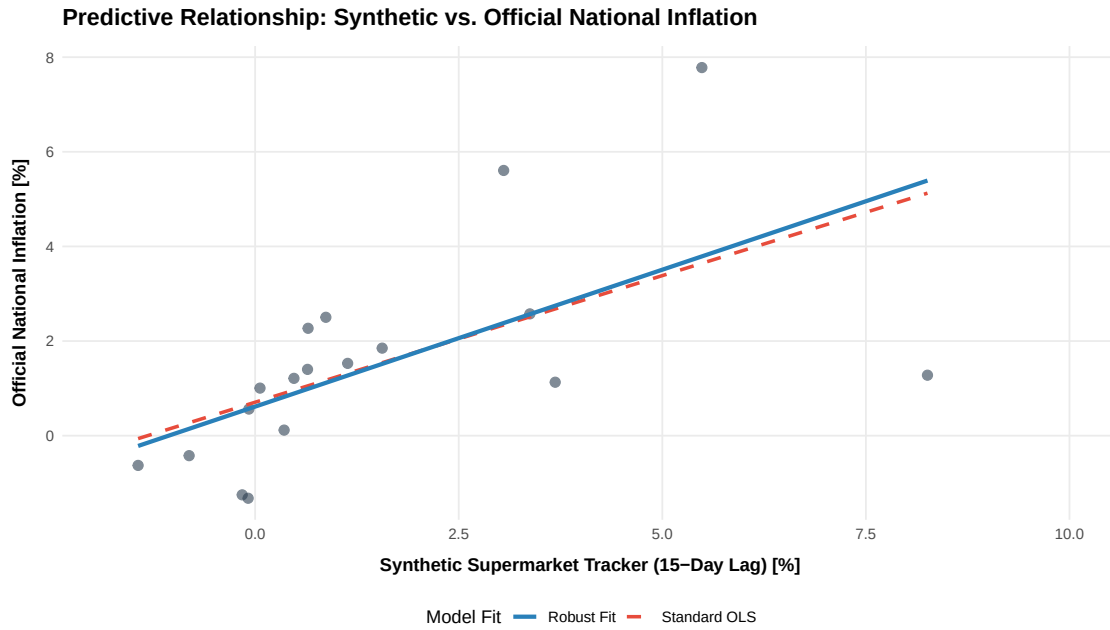


Figure 3: Relationship between 15-day lagged synthetic inflation and official “core-5” inflation. Includes OLS and Iteratively Reweighted Least Squares lines.

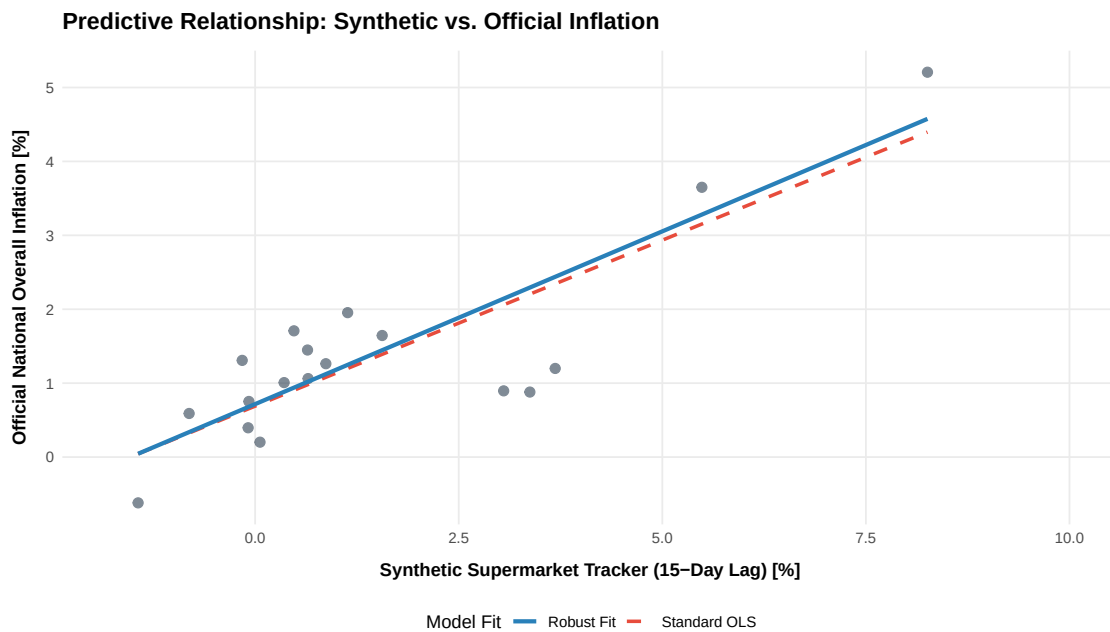


Figure 4: Relationship between 15-day lagged synthetic inflation and official overall inflation, full 12 categories. Includes OLS and Iteratively Reweighted Least Squares lines.

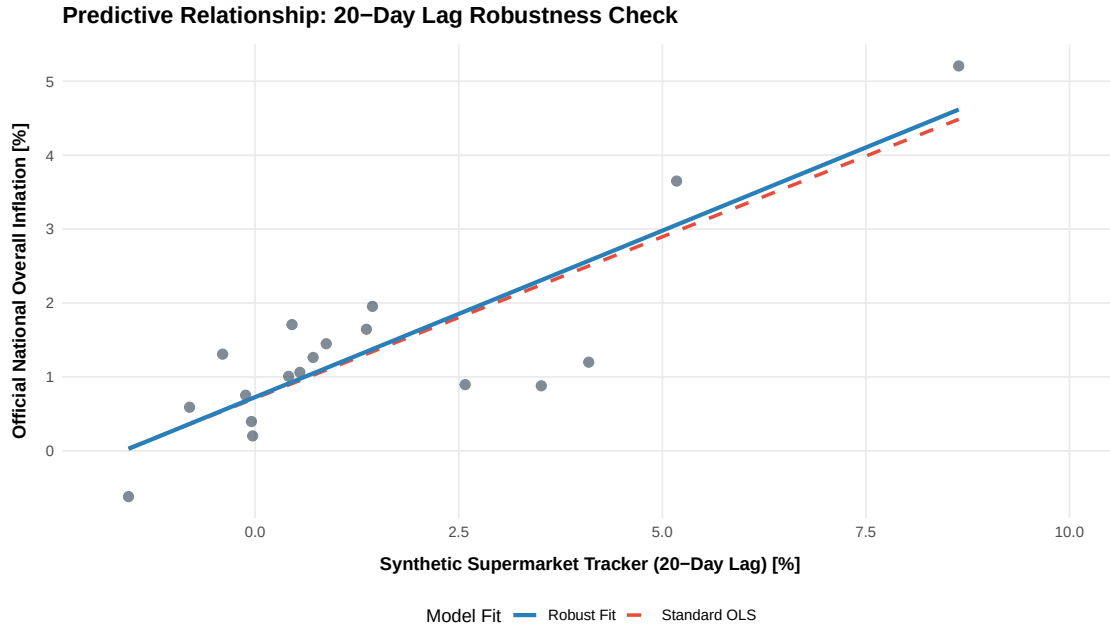


Figure 5: Relationship between 20-day lagged synthetic inflation and official overall inflation, full 12 categories. Includes OLS and Iteratively Reweighted Least Squares lines.

All three scatter plots suggest a positive linear relationship between lagged synthetic inflation and realized official inflation. Interestingly, synthetic inflation is more strongly correlated with the overall inflation (full 12 categories) than with “core-5” inflation.

Table 4: Robust Regression Estimates: Predicting Official Overall Inflation

Predictor Variables	Dependent Variable: Official National Overall Inflation	
	Model 1 (15-Day Lag)	Model 2 (20-Day Lag)
Intercept	0.716*** (0.211)	0.726*** (0.204)
Synthetic_Lag_15	0.467*** (0.075)	
Synthetic_Lag_20		0.450*** (0.071)
Model Diagnostics		
Observations	18	18
Residual Std. Error	0.722	0.808

*Note: Robust standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$*

Table 4 depicts two regression models with official overall inflation as the dependent variable and lagged synthetic inflation (15 and 20-day lags) as the independent variables. A one percentage point increase in synthetic inflation is associated with a 0.47 percentage point increase at the 15-day horizon and a 0.45 percentage point increase at the 20-day horizon.

4 Discussion

The synthetic index closely mirrors both the equivalent “core-5” CPI measure and overall CPI. CPI correlation coefficients range from 0.97 to 0.99, and MADs range from 2.3 to 2.8 index points (except for when compared to the overall CPI, which is 6). Likewise, although the relationship is not as strong, month-over-month synthetic inflation matches official inflation quite well. Correlation coefficients range from 0.70 to 0.80, and MADs are around 1.20 to 1.53 percentage points.

The very limited data also appears to suggest some degree of predictive power of the MoM (measured daily) synthetic inflation tracker on subsequent inflation. Surprisingly, the relationship is strongest between lagged synthetic inflation and official overall inflation (full 12 categories). A one percentage point increase in 15-day lagged synthetic inflation is associated with a 0.47 percentage point increase in overall inflation; this result is significant at the 0.01 level and also holds up at a 20-day lag. The coefficients of 0.47 and 0.45 (15-day and 20-day lag, respectively) are structurally consistent given that my index covers 49% of the overall CPI. It should be noted that with so few observations, the predictive power of the synthetic CPI may significantly change as time progresses.

Given the severity of the recent crisis, inflation will likely remain on the minds of many Bolivians even as it continues to cool. For the Bolivian government to deliver on its promise of low inflation, it must rein in fiscal deficits (and debt monetization), continue with the elimination of the fuel subsidy, and pursue growth-promoting policies such as financial and trade liberalization. A real-

time measure of the variable of interest not only allows policymakers to see the instantaneous impacts of new legislation but also allows them to preempt inflationary or deflationary pressure with countercyclical fiscal and/or monetary policy.

5 Appendix Tables

Table 5: INE Weights by Division

Division	Weight
Food and Non-Alcoholic Beverages	27.06
Alcoholic Beverages and Tobacco	0.88
Clothing and Footwear	7.56
Housing and Basic Services	8.56
Furniture, Household Goods and Services	6.08
Health	3.55
Transport	9.07
Communications	5.43
Recreation and Culture	6.22
Education	4.07
Food Consumed Away from Home	13.95
Miscellaneous Goods and Services	7.55

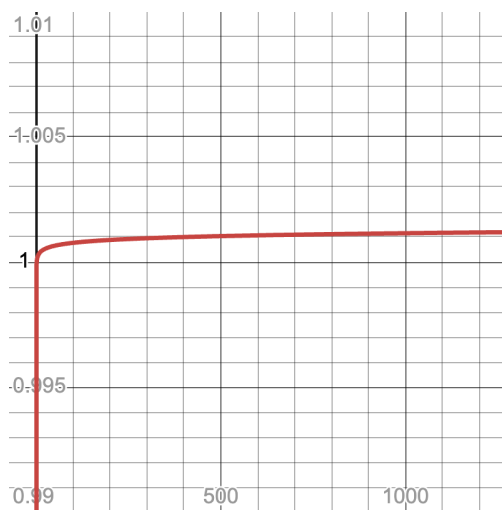


Figure 6: Geometric mean response to a single outlier price relative. Here, x is the outlier, and $y = (x \cdot 1)^{1/6000}$ is the geometric mean of 6,000 price relatives (assuming all other relatives stay at 1).

Table 6: Price Relative Summary Statistics by Subcategory

Subcategory	N	Min	Median	Mean	Geom Mean	Max	SD
Mueblería General	2156	0.7419	1	1.0026	1.0017	1.5657	0.0465
Carnes al Vacio Externas	12745	0.8485	1	1.0013	1.0012	1.4615	0.0143
Derivados Hipermaxi	6978	1.0000	1	1.0012	1.0011	1.4211	0.0154
Carne de Res	64652	0.7500	1	1.0011	1.0010	1.8800	0.0135
Menaje Plástico	115728	0.4857	1	1.0014	1.0009	3.5000	0.0368
Crema de Leche	12978	0.5077	1	1.0011	1.0009	1.9697	0.0222
Carnes Procesadas	16684	0.7987	1	1.0008	1.0008	1.3913	0.0128
Fraccionados Hipermaxi	36252	0.9231	1	1.0009	1.0007	1.9643	0.0185
Churrasquería	59680	0.5000	1	1.0009	1.0007	2.5000	0.0236
Quesos	242602	0.4798	1	1.0009	1.0007	2.3800	0.0184
Limpieza de Zapatos	50626	0.6061	1	1.0008	1.0007	1.5152	0.0151
Bioseguridad	2534	0.7500	1	1.0015	1.0007	4.0000	0.0621
Café y Sucedáneos	698100	0.0989	1	1.0008	1.0007	12.2923	0.0250
Cuidado del Adulto	61669	0.0843	1	1.0012	1.0007	10.9485	0.0483
Chocolates y Caramelos	595898	0.1301	1	1.0008	1.0007	6.2135	0.0200
Dulces de Leche y Manjares	23450	0.6106	1	1.0008	1.0007	1.5100	0.0183
Panadería Especialidades Hipermaxi	19698	0.6667	1	1.0009	1.0007	2.3568	0.0259
Golosinas - Varias	447053	0.0942	1	1.0009	1.0006	14.0000	0.0350
Accesorios del Hogar	494396	0.2937	1	1.0008	1.0006	3.5676	0.0212
Cuidado del Bebé	306161	0.1421	1	1.0009	1.0006	9.4872	0.0302
Yogures	212978	0.5263	1	1.0007	1.0006	1.9667	0.0161
Pañales	171106	0.5019	1	1.0010	1.0006	2.0729	0.0296
Comidas en Conserva	265738	0.1000	1	1.0008	1.0006	10.9000	0.0283
Salsas y Extracto de Tomate	97412	0.0101	1	1.0019	1.0006	108.4722	0.3461
Refrescos en Polvo	102671	0.1231	1	1.0007	1.0006	10.0000	0.0316
Menaje Vidrio/Cristal	196780	0.4444	1	1.0008	1.0006	1.8889	0.0218
Galletas	609473	0.1071	1	1.0008	1.0006	9.7391	0.0252
Mascotas	86016	0.5684	1	1.0007	1.0006	1.6792	0.0133
Verduras en Conserva	240082	0.1250	1	1.0007	1.0006	8.2000	0.0224
Articulos de Cocina	185295	0.4615	1	1.0007	1.0006	2.5714	0.0215
Nutrición Infantil	537155	0.4286	1	1.0007	1.0006	2.7000	0.0177
Fiambres Hipermaxi	148153	0.6014	1	1.0007	1.0005	1.7143	0.0157
Insecticidas	62265	0.4494	1	1.0008	1.0005	2.8000	0.0230
Kids	143479	0.4083	1	1.0007	1.0005	2.6000	0.0219
Básicos	888463	0.1000	1	1.0007	1.0005	10.8621	0.0220

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Table 6: Price Relative Summary Statistics by Subcategory (Continued)

Subcategory	N	Min	Median	Mean	Geom Mean	Max	SD
Cuidado de la Piel	1781232	0.1188	1	1.0008	1.0005	10.1000	0.0244
Cuidado Familiar	291479	0.0855	1	1.0010	1.0005	9.0851	0.0376
Carnes Congeladas	306477	0.5714	1	1.0006	1.0005	1.9014	0.0160
Carne de Pollo	30112	0.0415	1	1.0015	1.0005	23.3582	0.1308
Mantequillas y Margarinas	72925	0.5930	1	1.0006	1.0005	1.8182	0.0148
Sandwiches Delifresh	1834	1.0000	1	1.0005	1.0005	1.0917	0.0066
Licores	554238	0.0857	1	1.0006	1.0005	12.3529	0.0233
Postres Delifresh	3682	1.0000	1	1.0005	1.0005	1.1538	0.0080
Cuidado Bucal	458471	0.4167	1	1.0008	1.0005	3.0000	0.0261
Helados, Cassattas y Picoleses	300387	0.1928	1	1.0006	1.0005	5.5333	0.0158
Condimentos y Legumbres	639363	0.0996	1	1.0007	1.0005	10.9000	0.0270
Panadería Externa	195551	0.6500	1	1.0006	1.0005	1.6818	0.0131
Frutas en Conserva	120978	0.6849	1	1.0006	1.0005	1.8421	0.0138
Baño General	13871	0.5128	1	1.0007	1.0005	2.3810	0.0265
Articulos de Bebé	10585	0.7273	1	1.0007	1.0005	2.2500	0.0223
Vinos y Champagne	653269	0.1000	1	1.0006	1.0005	11.0000	0.0250
Repostería	371584	0.1000	1	1.0007	1.0005	10.7778	0.0257
Cuidado del Cabello	1334642	0.1042	1	1.0007	1.0005	12.0000	0.0213
Fiambres Externos	406192	0.4227	1	1.0005	1.0005	2.3659	0.0113
Pastelería Hipermáxi	34308	0.6644	1	1.0006	1.0005	2.0000	0.0161
Aderezos	400279	0.1020	1	1.0006	1.0005	10.0000	0.0244
Snack	562055	0.1000	1	1.0006	1.0005	10.0000	0.0206
Masas Típicas Hipermáxi	8760	0.8000	1	1.0006	1.0005	1.8085	0.0185
Limpieza de Living y Ropero	349022	0.1193	1	1.0007	1.0005	10.0000	0.0288
Dietéticos y Saludables	400495	0.1000	1	1.0006	1.0005	10.8182	0.0212
Masas y Comidas Congeladas	169769	0.6583	1	1.0005	1.0005	2.0308	0.0131
Pastelería Externa	124926	0.6933	1	1.0005	1.0005	1.7143	0.0103
Otras Carnes	2758	0.7258	1	1.0006	1.0004	1.2317	0.0159
Limpieza de la Ropa	568936	0.0862	1	1.0006	1.0004	9.5075	0.0264
Ferretería	77944	0.0976	1	1.0006	1.0004	8.7857	0.0331
Mascotas General	24410	0.7143	1	1.0005	1.0004	2.8000	0.0177
Gaseosas	195902	0.6667	1	1.0005	1.0004	1.5000	0.0118
Limpieza de Baños	360432	0.4318	1	1.0005	1.0004	2.6842	0.0171
Cervezas	107346	0.6102	1	1.0005	1.0004	1.6667	0.0154
Jugos, Néctar	225845	0.3333	1	1.0005	1.0004	3.0000	0.0147

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Table 6: Price Relative Summary Statistics by Subcategory (Continued)

Subcategory	N	Min	Median	Mean	Geom Mean	Max	SD
Leches	199984	0.2632	1	1.0007	1.0004	4.2857	0.0277
Frutas	52833	0.3333	1	1.0014	1.0004	2.6667	0.0471
Postres	27245	0.6190	1	1.0007	1.0004	1.7333	0.0252
Cotillón	81511	0.6522	1	1.0005	1.0004	2.0000	0.0142
Ofertas Especiales	11624	0.8326	1	1.0005	1.0004	1.3367	0.0182
Automotriz	9886	0.8125	1	1.0005	1.0004	1.9143	0.0165
Energizantes	48031	0.7258	1	1.0004	1.0004	1.5556	0.0128
Limpieza de Cocina	266112	0.1556	1	1.0006	1.0004	6.0000	0.0240
Menaje Porcelana/Cerámica	25185	0.6824	1	1.0005	1.0003	2.0886	0.0186
Mermel, Miele	246511	0.5778	1	1.0004	1.0003	1.8868	0.0107
Pulpas y Frutas Congeladas	71180	0.4103	1	1.0004	1.0003	2.3113	0.0129
Panadería Hipermaxi	56176	0.1345	1	1.0005	1.0003	1.7538	0.0184
Bebidas Varias	156261	0.5676	1	1.0004	1.0003	2.0000	0.0126
Verduras	104967	0.1375	1	1.0015	1.0003	9.1818	0.0583
Productos Importados	75711	0.1877	1	1.0012	1.0003	5.3267	0.0629
Masas Típicas Externas	59041	0.8378	1	1.0003	1.0003	1.3761	0.0067
Gourmet	24497	0.9420	1	1.0003	1.0003	1.6667	0.0092
Aguas	112034	0.6250	1	1.0003	1.0003	1.6875	0.0093
Bijoutería/Cuidado Personal	109532	0.4778	1	1.0005	1.0003	2.2000	0.0244
Diferenciales Usa	427	1.0000	1	1.0003	1.0002	1.1124	0.0054
Varios	2116	1.0000	1	1.0003	1.0002	1.4118	0.0096
Isotónicos	107885	0.6667	1	1.0003	1.0002	1.5000	0.0145
Cigarrillos	125781	0.6763	1	1.0002	1.0002	1.3125	0.0080
Hortalizas Delifresh	17307	0.5882	1	1.0003	1.0002	1.5517	0.0134
Electrodoméstico Cuidado Personal	2275	0.6455	1	1.0003	1.0002	1.6349	0.0179
Librería	7156	0.6000	1	1.0003	1.0002	1.9444	0.0184
Menaje Metálico	3203	0.5000	1	1.0005	1.0002	2.4667	0.0319
Hielos	4701	0.9333	1	1.0001	1.0001	1.1429	0.0039
Camping General	11152	0.4992	1	1.0003	1.0001	2.0000	0.0232
Huevos	67342	0.6731	1	1.0001	1.0000	1.4423	0.0143
Bolsos/Carteras/Maletas	2449	0.7317	1	1.0000	1.0000	1.2000	0.0080
Accesorios Del Hogar	1	1.0000	1	1.0000	1.0000	1.0000	NA
Artículos Bebé	95	1.0000	1	1.0000	1.0000	1.0000	0.0000
Aseo Del Bebé	1	1.0000	1	1.0000	1.0000	1.0000	NA

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Table 6: Price Relative Summary Statistics by Subcategory (Continued)

Subcategory	N	Min	Median	Mean	Geom Mean	Max	SD
Chocol. y Caramelos	3	1.0000	1	1.0000	1.0000	1.0000	0.0000
Cuidado De La Piel	1	1.0000	1	1.0000	1.0000	1.0000	NA
Cuidado Del Cabello	1	1.0000	1	1.0000	1.0000	1.0000	NA
Dermatológicos	290	1.0000	1	1.0000	1.0000	1.0000	0.0000
Jugueteria Marcas	19	1.0000	1	1.0000	1.0000	1.0000	0.0000
Juguetería Niña/Bebé	390	1.0000	1	1.0000	1.0000	1.0000	0.0000
Juguetería Niño/Bebé	10	1.0000	1	1.0000	1.0000	1.0000	0.0000
Limpieza De La Ropa	1	1.0000	1	1.0000	1.0000	1.0000	NA
Limpieza De Living y Roperos	2	1.0000	1	1.0000	1.0000	1.0000	0.0000
Ropa Dama/Junior	402	1.0000	1	1.0000	1.0000	1.0000	0.0000
Ropa Niño/Adolescente	344	1.0000	1	1.0000	1.0000	1.0000	0.0000
Tracto Alimentario y Metabolismo	38	1.0000	1	1.0000	1.0000	1.0000	0.0000
Electrodoméstico/Línea Blanca-Hogar	12468	0.4664	1	1.0001	1.0000	1.8571	0.0151
Productos Americanos	434511	0.1023	1	1.0006	0.9999	9.1633	0.0464
Regalos/Adornos	19354	0.5000	1	1.0003	0.9999	2.0000	0.0300
Artículos para Bebé	1155	0.6842	1	0.9997	0.9997	1.0000	0.0093
Bebidas Usa	81010	0.1000	1	1.0015	0.9995	5.0201	0.0778
Cocina General	87552	0.4000	1	0.9998	0.9995	2.0339	0.0258
Electrodomésticos Cuidado Personal	1549	0.4545	1	0.9996	0.9995	1.0000	0.0139
Living y Dormitorio	92734	0.2817	1	0.9997	0.9994	2.2414	0.0224

Table 9: Complete Subcategory List by Product Volume

Subcategory	Observations
Cuidado de la Piel	1,800,677
Cuidado del Cabello	1,336,777
Básicos	884,422
Café y Sucedáneos	695,349
Vinos y Champagne	649,930
Condimentos y Legumbres	636,724
Galletas	607,444
Chocolates y Caramelos	593,662
Limpieza de la Ropa	566,359

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Table 7: Expenditure Categories and Values

Category	Value
Food Consumed Away from Home	5
Food and Beverages Consumed Away from Home	7
Food and Non-Alcoholic Beverages	14,931
Alcoholic Beverages and Tobacco	1,772
Miscellaneous Goods and Services	5,726
Communications	24
Education	54
Furniture, Household Goods and Services	4,040
Clothing and Footwear	60
Recreation and Culture	1,535
Restaurants and Hotels	4
Health	122
Transport	57
Housing and Basic Services	41

Table 8: Summary Statistics: Time Between Price Observations

Statistic	Value
<i>Panel A: Distribution of Observation Gaps (Days)</i>	
Minimum	1.000
1st Quartile	1.000
Median	1.000
Mean	1.135
3rd Quartile	1.000
Maximum	636.000
<i>Panel B: Tail Risk (Percentiles)</i>	
50th Percentile	1.000
75th Percentile	1.000
90th Percentile	1.000
95th Percentile	1.000
99th Percentile	2.000
<i>Panel C: Churn Frequency</i>	
Observations with Gap > 1 Day	1.21%

Notes: Table displays the empirical distribution of days between successfully scraped price observations for individual product SKUs across all surveyed cities.

Table 9: Complete Subcategory List by Product Volume
(Continued)

Subcategory	Observations
Snack	559,747
Licores	551,504
Nutrición Infantil	535,377
Accesorios del Hogar	493,899
Cuidado Bucal	456,611
Golosinas - Varias	445,672
Productos Americanos	433,773
Fiambres Externos	404,715
Dietéticos y Saludables	402,618
Aderezos	398,756
Repostería	370,956
Limpieza de Baños	360,144
Limpieza de Living y Ropero	348,292
Cuidado del Bebé	305,419
Carnes Congeladas	305,036
Helados, Cassattas y Picoleses	299,267
Cuidado Familiar	290,566
Limpieza de Cocina	264,906
Comidas en Conserva	264,748
Mermel, Miele	248,437
Quesos	241,689
Verduras en Conserva	239,229
Mascotas	230,052
Jugos, Néctar	225,002
Yogures	212,092
Menaje Vidrio/Cristal	204,709
Leches	199,325
Gaseosas	195,050
Panadería Externa	194,973
Artículos de Cocina	186,621
Pañales	170,455
Masas y Comidas Congeladas	169,142
Bebidas Varias	155,777

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Table 9: Complete Subcategory List by Product Volume
(Continued)

Subcategory	Observations
Fiambres Hipermaxi	147,403
Kids	143,045
Librería	141,491
Cigarrillos	125,317
Pastelería Externa	124,384
Frutas en Conserva	120,466
Menaje Plástico	116,576
Cotillón	114,703
Aguas	111,649
Bijouteria/Cuidado Personal	110,458
Mascotas General	108,621
Isotónicos	107,422
Cervezas	107,007
Ferretería	105,344
Verduras	104,506
Refrescos en Polvo	102,804
Salsas y Extracto de Tomate	97,128
Living y Dormitorio	93,113
Cocina General	87,485
Temporada Navidad	81,211
Bebidas Usa	80,837
Productos Importados	75,241
Mantequillas y Margarinas	72,657
Pulpas y Frutas Congeladas	70,976
Huevos	67,046
Carne de Res	64,295
Churrasquería	64,222
Insecticidas	61,994
Cuidado del Adulto	61,457
Masas Típicas Externas	58,808
Panadería Hipermaxi	55,879
Frutas	52,593
Limpieza de Zapatos	50,406

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Table 9: Complete Subcategory List by Product Volume
(Continued)

Subcategory	Observations
Energizantes	47,729
Automotriz	47,003
Regalos/Adornos	43,944
Juguetería Niño/Bebé	43,709
Fraccionados Hipermáxi	36,206
Pastelería Hipermáxi	34,164
Temporada	30,547
Carne de Pollo	30,019
Postres	27,175
Menaje Porcelana/Cerámica	25,166
Gourmet	24,406
Dulces de Leche y Manjares	23,355
Juguetería Niña/Bebé	22,469
Panadería Especialidades Hipermáxi	20,235
Juguetería Didáctica/Temporada	20,153
Hortalizas Delifresh	17,276
Carnes Procesadas	16,606
Baño General	13,829
Camping General	13,374
Crema de Leche	12,935
Electrodoméstico/Línea Blanca-Hogar	12,866
Carnes al Vacío Externas	12,705
Temporada Halloween	11,978
Ofertas Especiales	11,574

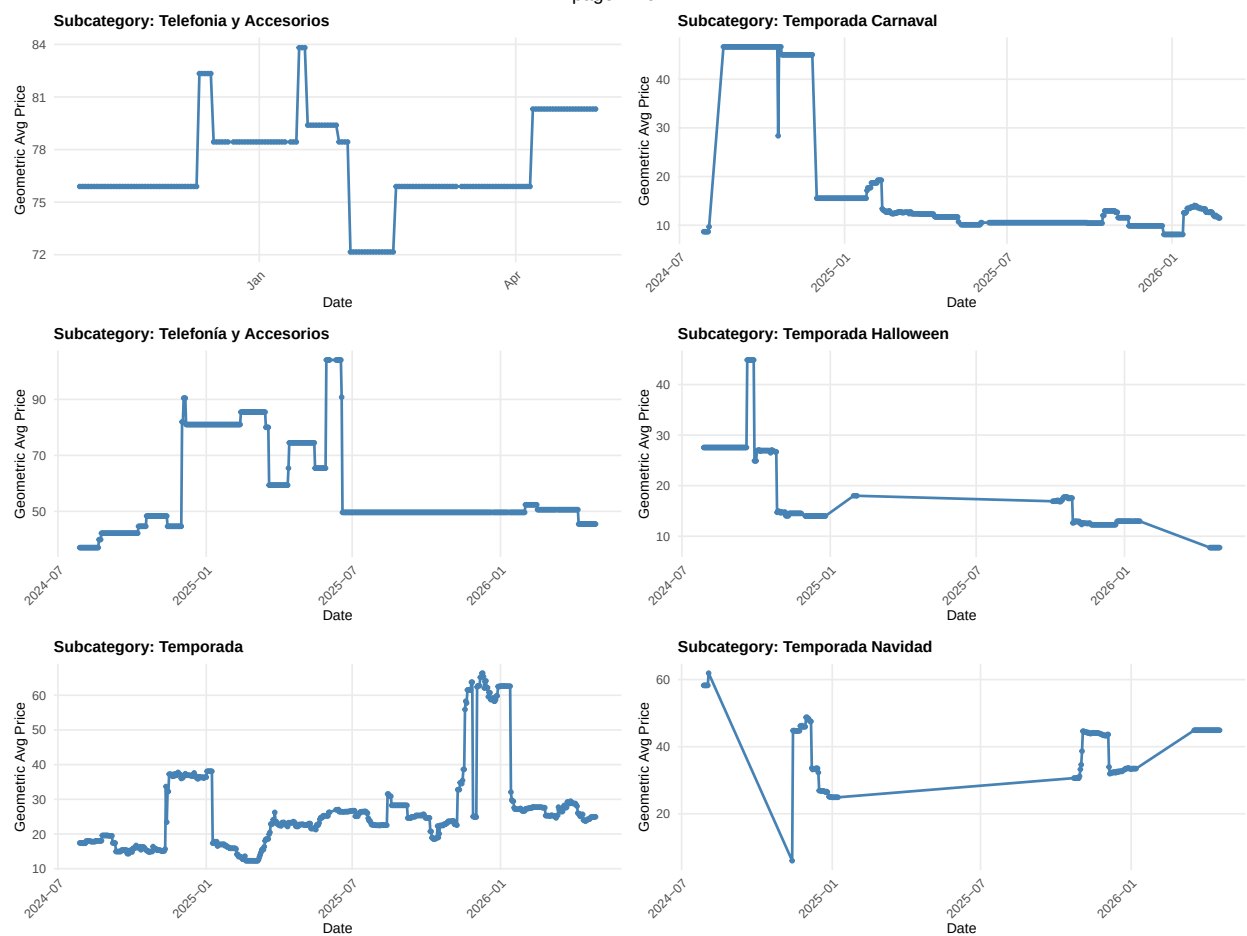


Figure 7: Subcategory geometric average price series.

References

- [1] Instituto Nacional de Estadística (INE). Gráficos ipc, 2026. Accessed May 6, 2026.
- [2] International Monetary Fund. Imf datamapper: Gross domestic product, constant prices, annual percent change—bolivia, 2026. Accessed May 6, 2026.
- [3] Instituto Nacional de Estadística (INE). Pib trimestral—gráfico y datos, 2026. Accessed May 6, 2026.
- [4] World Bank. Macro poverty outlook: Bolivia. World Bank report, 2021. Published August 24, 2021. Accessed May 6, 2026.
- [5] Alberto Cavallo. Online and official price indexes: Measuring argentina’s inflation. *Journal of Monetary Economics*, 60(2):152–165, 2013. Accessed May 6, 2026.
- [6] Mauricio Foronda. precios: Variaciones de precios en bolivia. <https://github.com/mauforonda/precios>, 2026. GitHub repository.